



ANTARIKHYA

अन्तरिक्ष प्रतिष्ठान नेपाल



Slippers2Sat-2 (S2S-2): Camera Mission

S2S-2 Team

Kathmandu, Nepal



ANTARIKSHA

अन्तरिक्ष प्रतिष्ठान नेपाल



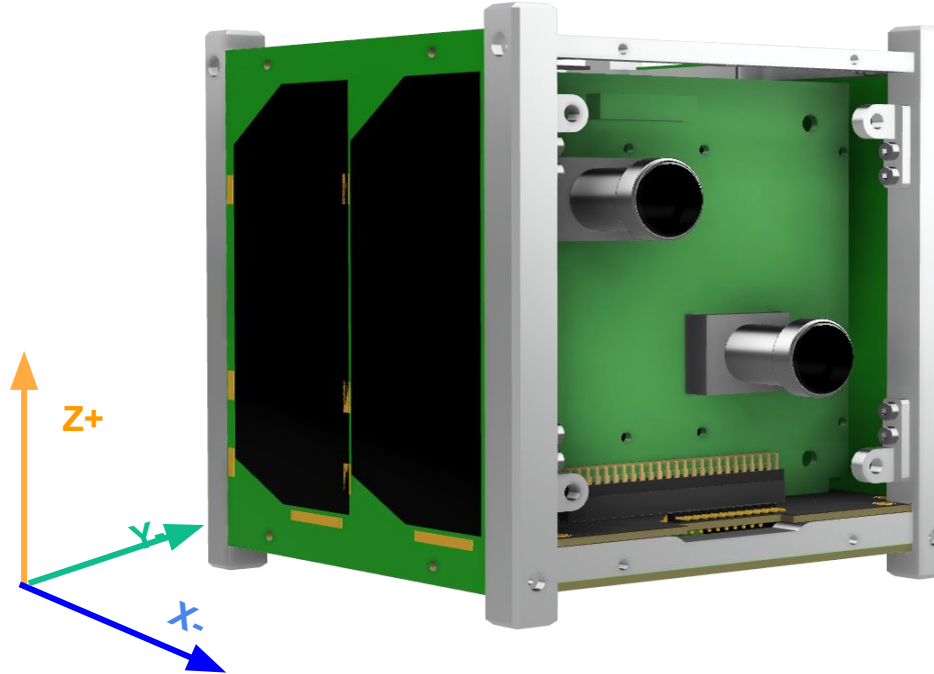
MSN2: Camera Mission (CAM)



ANTARIKSHA

अन्तरिक्ष प्रतिष्ठान नेपाल

Camera (CAM)





ANTARIKSHA

अन्तरिक्ष प्रविष्टान नेपाल

Introduction



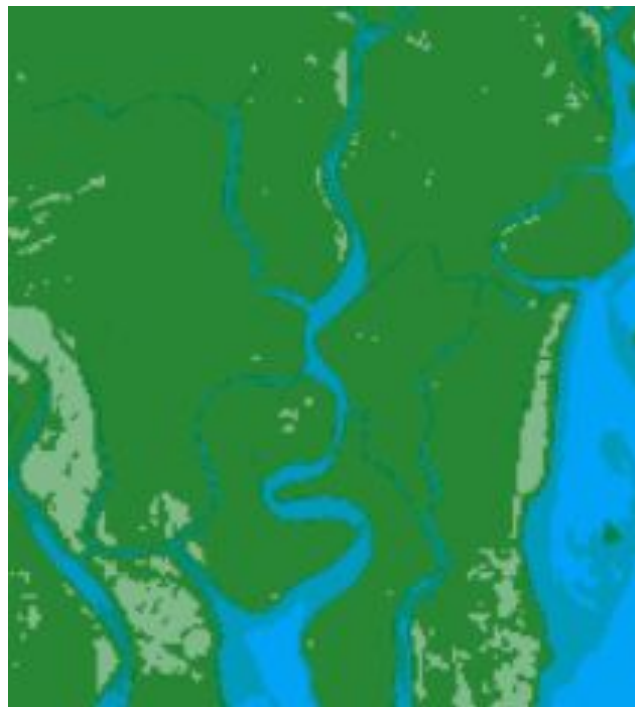
- Detect and monitor slight changes in water content of the water bodies.
- Used in flood, drought monitoring.

NDWI:

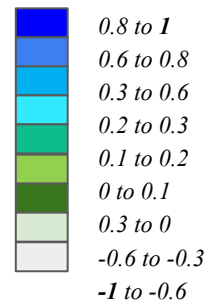
$$(\text{rgreen} - \text{rNIR})/(\text{rgreen} + \text{rNIR})$$

rgreen: reflectance of green band

rNIR: the reflectance NIR bands



Level of water source



45.01 Km x 28.07 Km

4

Representation of NDWI from Landsat 8 satellite imagery

[Southern Bay of Bengal; taken from landviewer open source portal]

- The NDWI value ranges from 1 to -1 and zero is the threshold



ANTARIKSHA

अन्तरिक्ष प्रतिष्ठान नेपाल

Introduction



RGB image of earth.

Combination of three spectral bands; **Red, Green and blue.**

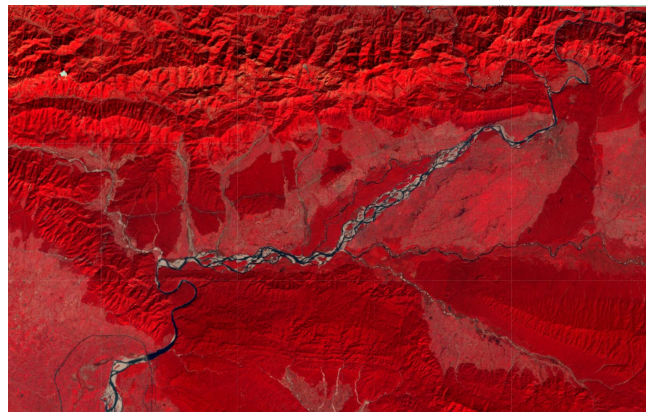
These primary colors can be **mixed to create any color** and give **true colour image**.



Longitude: 84.12750068246642 Latitude: 27.55730418747308 A: 81 km x 51.5 km

NIR image of the earth.

Water mixed with sediments reflect light and appear blue.
clear water is black.



Longitude: 84.12750068246642 Latitude: 27.55730418747308 A: 81 km x 51.5 km



ANTARIKSHA

अन्तरिक्ष प्रविष्टान नेपाल

Statement and Objective



- Mission Statement
 - The Camera mission aims to capture picture of earth in two spectrum (namely; **visible** and **near- infrared**) visualizing surfaces with water bodies using two cameras; RGB and NIR
- Mission objectives
 - To take terrestrial picture of the earth containing vegetation, landforms and water bodies
 - To use captured images to determine Normalized Difference Water Index (NDWI).

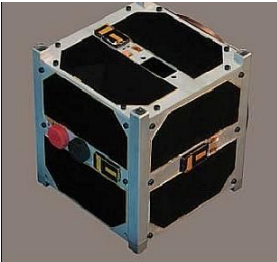
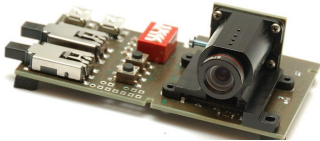


ANTARIKSHA

अन्तरिक्ष प्रतिष्ठान नेपाल

Past Mission (1)



SN	SATELLITE (CUBESAT)	IMAGE SENSOR USED	MASS (Avg.)	FORM FACTOR	CAMERA MISSION	STATUS
1	EST Cube1 	Aptina MT9V011 	1.3 kg	1U	Monitor the tether deployment of ESTCube-1 and capture images of Earth	Mission complete (2023)
2	FACSAT-1 	Aptina MT9T031 	-	3U	Earth Observation (EO) and use of a ground-based system: the Ultrahigh Frequency (UHF) transceiver.	Decayed; Satellite has re-entered

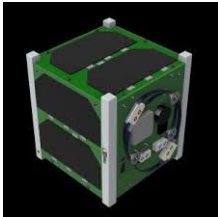


ANTARIKSHA

अन्तरिक्ष प्रतिष्ठान नेपाल

Past Mission(2)



3	Missat-1		COMedia C6820		1.3 kg	1U	Capture images of earth and send those images to the ground station at the University of Mississippi while in orbit	—
4	Birds - 4		OV5642		1.3 kg	1U	Obtain visual data on specific targets or area of interest.	—
5	MUNAL		OV5640		1.038 kg	1U	Capture high resolution RGB and NIR image of earth for vegetation mapping.	Not launched yet.



ANTARIKSHA

अन्तरिक्ष प्रतिष्ठान नेपाल

Requirements (CAM)



ID	System Requirement	Design Requirement	Verification Requirements
SR1.1	Mission shall take RGB image of earth	CAM MCU responds to OBC signal and command via communication protocol	Shall send acknowledgement data from CAM MCU to OBC MCU via UART
		Shall be RGB passive imager	Shall have IR filter coating in lens
		Shall have image compression to reduce the size of data	Shall have JPEG compression



ANTARIKSHA

अन्तरिक्ष प्रतिष्ठान नेपाल

Requirements (CAM)



ID	System Requirement	Design Requirement	Verification Requirements
SR1.1	Mission shall take RGB image of earth	Camera sensor shall have 1280x720 image resolution	Shall have HD resolution
		Shall have ground swath of less than 100 Km	FOV > 10°
		Shall have a focal length of more than 12 mm	Shall give Ground Sampling Distance less than 100 meters



ANTARIKSHA

अन्तरिक्ष प्रविष्टान नेपाल

Requirements (CAM)



ID	System Requirement	Design Requirement	Verification Requirements
SR1.1	Mission shall take RGB image of earth	Camera sensor shall have 640x480 image resolution	Shall have VGA resolution
		Shall have ground swath of less than 100 Km	$10^{\circ} < \text{FOV} < 15^{\circ}$
		Shall have a focal length of more than 12 mm	Shall give Ground Sampling Distance less than 150 meters



Requirements (CAM)



ID	System Requirement	Design Requirement	Verification Requirement
SR 1.2	Mission shall take NIR image of the earth	Secondary Camera (NIR) system shall respond to OBC signal and command via communication protocol	Shall send acknowledgement data from CAM MCU to OBC MCU
		Shall be NIR passive imager	Shall have IR pass filter coating in lens
		Shall have an image compression algorithm to reduce the size of data	Shall have JPEG compression



ANTARIKSHA

अन्तरिक्ष प्रविष्टान नेपाल

Requirements (CAM)



ID	System Requirement	Design requirement	Verification Requirement
SR 1.2	Mission shall take NIR image of the earth	Shall have 1280x720 image resolution	Shall have HD resolution
		Shall have focal length more than 12 mm	Shall give Ground Sampling Distance of less than 100m meter
		Shall have ground swath less than 100 Km	FOV > 10°

FOV : Field of view

**Estimated*



ANTARIKSHA

अन्तरिक्ष प्रतिष्ठान नेपाल

Requirements (CAM)



ID	System Requirement	Design Requirement	Verification Requirements
SR1.1	Mission shall take NIR image of earth	Camera sensor shall have 640x480 image resolution	Shall have VGA resolution
		Shall have ground swath of less than 100 Km	$10^{\circ} < \text{FOV} < 15^{\circ}$
		Shall have a focal length of more than 12 mm	Shall give Ground Sampling Distance less than 150 meters



ANTARIKSHA

अन्तरिक्ष प्रतिष्ठान नेपाल

Requirements (CAM)



ID	System Requirement	Design Requirement	Verification Requirement
SR 1.3	System Shall be within power limitation.	Shall consume less than 2% of power generated	Shall consume $< 2.3 \text{ W}^*$

*Estimated



ANTARIKSHA

अन्तरिक्ष प्रविष्टान नेपाल

Requirements (CAM)



ID	System Requirement	Design Requirement	Verification Requirement
SR 1.3	System shall be within design requirement	Shall consume < 1% of the total average energy generated	Shall consume < 11.736 mWh*
		System shall have not more than 17.5% of the total mass	Shall weight < 230 gm

**Estimated (per orbit cycle; operation time = 10 sec)*



ANTARIKSHA

अन्तरिक्ष प्रतिष्ठान नेपाल

Requirements (CAM)



ID	System Requirement	Design Requirement	Verification Requirement
SR 1.4	System shall be within design limitations	Payload shall fit inside structure	payload shall be limited within 90mm x 86 mm board
		Camera lens shall fit within structure	Shall be <40mm in full length; (-X axis) protruding from its MSN board Camera lens must have $\varnothing < 40$ mm
SR 1.5	Payload shall withstand space environment	Shall withstand extreme temperature range	Shall operate in the range -30°C to 55°C range



ANTARIKSHA

अन्तरिक्ष प्रविष्टान नेपाल



Block diagram

EPS OBC Board

FM_2

(4)

MUX

(4)

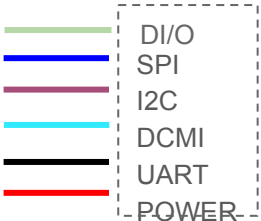
EPS OBC

(2)

ADCS

(1)

ADCS Board



CAM Mission Board

SDRAM

(4)

NIR MCU

(8)

(2)

NIR imager Sensor

3.3V

NIR_memory

(4)

OCP_2

(1)

RGB MCU

(8)

(2)

RGB Imager Sensor

3.3V

SDRAM

(4)

RGB_memory

(4)

OCP_2

3V3

(1)

OCP_4

5V

LDO

Power line

5V

MUX: Multiplexer
FM: Flash Memory
OCP: Overcurrent Protection
DCMI: Digital Camera Memory Interface
LDO: Low dropout Voltage Regulator



ANTARIKSHA

अन्तरिक्ष प्रतिष्ठान नेपाल

Optics: Component specs



Sensor Specs (OV5640)	Pixel size (μm)	$1.4 \mu\text{m} \times 1.4 \mu\text{m}$
	Vertical sensor size (mm)	$2738.4 \mu\text{m}$
	Horizontal sensor size (mm)	$3673.6 \mu\text{m}$
	Diagonal sensor size (mm)	$3315.74 \mu\text{m}$
Parameters	OpenMV H7 plus camera module from OpenMV Cam.	
Image sensor	CMOS	
Mount	M12	
Shutter method	Rolling Shutter	
Dimension	$35.56 \text{ mm} \times 18 \text{ mm} \times 10 \text{ mm}$	
Operating temperature range	$(-30^{\circ}\text{C} \text{ to } 70^{\circ}\text{C})$	



**ANTARIKSHA**

अन्तरिक्ष प्रविष्ठान नेपाल

Component selection: Lens



Company	Vision Dimension	Edmund optics C VIS-NIR Series #67-714	ArduCam CS2316ZM02
lens format	2/3"	2/3"	1/2.3"
Mount	M30.5x0.5 (C mount)	M25.5 x 0.50 (C mount)	CS mount
Focal length	16 mm	16 mm	16 mm
HFOV	for 1/4" 8.24°	for 1/4" 24.9 mm- 12.8°	13°
Dimension	ø 33.5mm, L 28.2mm, 60g	ø 33mm, L 40.50 mm, 48g	Ø30mm, L 29mm, 45g
Aperture	f/1.4	f/1.4 - f/1.6	f/1.2
Operating temperature		-20 to +60	-

 Considered



Lens: Parameter requirement



Parameters (Lens)	Requirement	ArduCam CS2316ZM02
Focal Length	>12 mm	16 mm
FOV	>10°	for 1/4", 13°
Aperture	>f/8	f/1.2
GS (km)	Less than 100 Km.	92 X 52 Km*
GSD(m)	<100m	72 km*
Weight (mm)	(< 50g)	45g
Mount	(M12) S-mount	CS mount



[11]

**Estimated, orbital height = 400 Km
refer to appendix.*

Comparative Analysis of System Requirement (1)



Requirement	Component Specification
Sensor have NIR imager	Equipped with IR filter
Payload shall fit inside 90mm x 86mm board.	Sensor dimension : 35.56 mm x 18 mm x 10 mm



ANTARIKSHA

अन्तरिक्ष प्रविष्टान नेपाल

Comparative Analysis of System Requirement (2)



Requirement	Component Specification
COM protocol from OBC MCU to CAM MCU via UART	Interface equipped with UART
Resolution of at least 1280x720	Supports upto QSXGA* (2592x1944)
Image compression algorithm	Supports JPEG image format
Tolerate space environment (temperature)	Provided range** : 30°C to 70°C

**(Quad Super XGA) ; A screen resolution of 2560x2048 pixels*

***stable image: 0°C to 50°C*

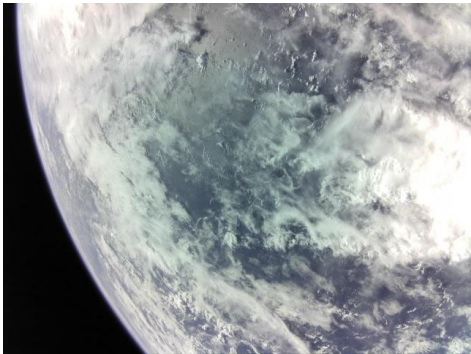
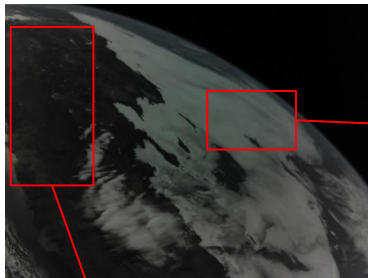


ANTARIKSHA

अन्तरिक्ष प्रतिष्ठान नेपाल

Mission Modes



Mission mode	Description		
	Operation mode 1	Operation mode 2	Operation mode 3
	Capture Low resolution 640x480(VGA) image only.	Capture HD resolution 1280x720 image only.	Image segmentation of captured respective image.
	 <i>640x480 Image of earth taken by Bird's-3 cubesat</i>	 <i>1280x960 Image of earth taken by Bird's-3 Cubesat.</i>	 <i>Image feature extraction.</i>



ANTARIKSHA

अन्तरिक्ष प्रतिष्ठान नेपाल

Mission Modes



Image mode				
RGB Image Size (bytes)	VGA (640x480)		HD (1280x720)	
	RAW	JPEG*	RAW	JPEG
	921600	115200	276400	345600
NIR Image Size (bytes)	307200	38400	921600	115200



ANTARIKSHA

अन्तरिक्ष प्रतिष्ठान नेपाल

Feasibility Test (Size)



Mission Name	Components	Volume	Compatibility
Digipeater (MSN1)	NXT_GEN_CUBUS	50x100x113mm ³	YES
CAM (MSN2)	Imaging Sensor	6x6x0.75 mm ³	YES
	Lens x2	$[\pi \times (15)^2 \times (22.5) \text{mm}^3$ $(\pi r^2 h)] * 2$	YES
ADCS (MSN3)	Magnetorquer x2	72x15x14mm ³	YES
	GPS Receiver	25.14x25.13x4.4mm ³	YES
EPDM (MSN4)	Magnetometer x4	25.4x25.4x7.1mm ³	YES



ANTARIKSHA

अन्तरिक्ष प्रतिष्ठान नेपाल

Feasibility Test (Weight)



Mission Name	Components	Mass	Compatibility
Digipeater (MSN1)	NXT_GEN_BUS	~0.476kgs	YES
CAM (MSN2)	Camera Sensor	0.017kgs	YES
	Lens	0.09kgs	YES
ADCS (MSN3)	Magnetorquer x2	0.03kgs	YES
	GPS Receiver	0.18kgs	YES
EPDM (MSN4)	Magnetometer x4	0.059kgs	YES



ANTARIKSHA

अन्तरिक्ष प्रतिष्ठान नेपाल

Feasibility Test (Power)



3SP E(30%) = 1173.6 mWh

3SP E(32%) = 1291.2 mWh

BUS/ Mission	Telemetry			Mission			
	Burner	Beacon	HK-Data	DP-M	ADCS	CAM	EPDM
EPS/OBC	1136.66	1136.66	1136.66	1136.66	1136.66	1136.66	1136.66
COM TX (Beacon)	OFF	49.50430556	OFF	OFF	OFF	OFF	OFF
COM TX (Downlink)	OFF	OFF	129.1416667	OFF	OFF	OFF	OFF
COM RX (Uplink)	OFF	24.288	24.288	OFF	24.288	24.288	24.288
DP-M TX (Downlink)	OFF	OFF	OFF	129.1416667	OFF	OFF	OFF
DP-M RX (Uplink)	OFF	OFF	OFF	24.288	OFF	OFF	OFF
ADCS	OFF	OFF	OFF	OFF	1.739444444	1.739444444	OFF
CAM	OFF	OFF	OFF	OFF	OFF	25.10866667	OFF
EPDM	OFF	OFF	OFF	OFF	OFF	OFF	130.35
Burner Circuit	33	OFF	OFF	OFF	OFF	OFF	OFF
GPS	OFF	0.092	0.092	0.092	0.092	0.092	0.092
Total Energy (mWh)	1169.660	1210.452	1290.090	1290.090	1163.196	1187.796	1291.390



Feasibility Test (Cost and Logistics)



Components	Unit	Cost	Logistic
OpenMV H7 plus module	2	\$89.00U SD	Digikey India
Arducam CS2316ZM02	2	\$55.0 USD	Edmund optics
Pixco Lens adapter	2	\$8.79	Amazon



ANTARIKSHA

अन्तरिक्ष प्रतिष्ठान नेपाल

Mission Success Criteria



Success Level	Criterion
Failure	No image is retrieved by the satellite
Minimum	Take RGB and NIR images in low resolution 640x480 (VGA) facing earth
Medium	Take RGB and NIR images in HD, 1280x720 resolution facing earth
Full	Image segmentation using Artificial Intelligence (machine Learning) for efficient image downlink
Extra	Camera mission operates without issue till S2S's End of Life (EoL)